

Limits. Part 2

January 22

In this session we are going to learn how to compute limits with a numerical approach and understand the main difference of computing the limit and just evaluate a function.

By evaluating a function, it answers to the question “*What is the function’s value at this point?*” and it just inserting a number directly to the algebraic expression. On the other hand, taking a Limit allows to answer to “*What value does the function approach as we get arbitrarily close to a point?*” And the function doesn’t even have to be defined at the point itself.

For example, let’s consider

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

If we evaluate the function at $x = 1$ we get $0/0$, which is undefined. Furthermore, since is $0/0$, this means that at $x = 1$ the function has a hole.

In order to know the value that the function is approaching when the domain goes to 1, let’s compute the limit using a numerical method. First let’s build a table with values that are near 1, for instance,

x	$f(x) = \frac{x^2 - 1}{x - 1}$
1.5	$\frac{(1.5)^2 - 1}{1.5 - 1} = \frac{2.25 - 1}{0.5} = \frac{1.25}{0.5} = 2.5$
1.3	$\frac{(1.3)^2 - 1}{1.3 - 1} = \frac{1.69 - 1}{0.3} = \frac{0.69}{0.3} = 2.3$
1.1	$\frac{(1.1)^2 - 1}{1.1 - 1} = \frac{1.21 - 1}{0.1} = \frac{0.21}{0.1} = 2.1$
1.01	$\frac{(1.01)^2 - 1}{1.01 - 1} = \frac{1.0201 - 1}{0.01} = \frac{0.0201}{0.01} = 2.01$
1.001	$\frac{(1.001)^2 - 1}{1.001 - 1} = \frac{1.002001 - 1}{0.001} = \frac{0.002001}{0.001} = 2.001$
1.000001	$\frac{(1.000001)^2 - 1}{1.000001 - 1} = \frac{1.000002000001 - 1}{0.000001} = \frac{0.000002000001}{0.000001} = 2.000001$

With this we can say that the “*limit*” tells us the value the function is trying to be at $x = 1$, even though it never actually reaches it there.

Class Activity

Compute the limit with the numerical approach of the following functions,

$$f(x) = \frac{1}{x} \quad x \rightarrow 0$$

$$g(x) = \frac{x^2 - 4}{x - 2} \quad x \rightarrow 2$$

$$h(x) = \frac{\sin(x)}{x} \quad x \rightarrow 0$$